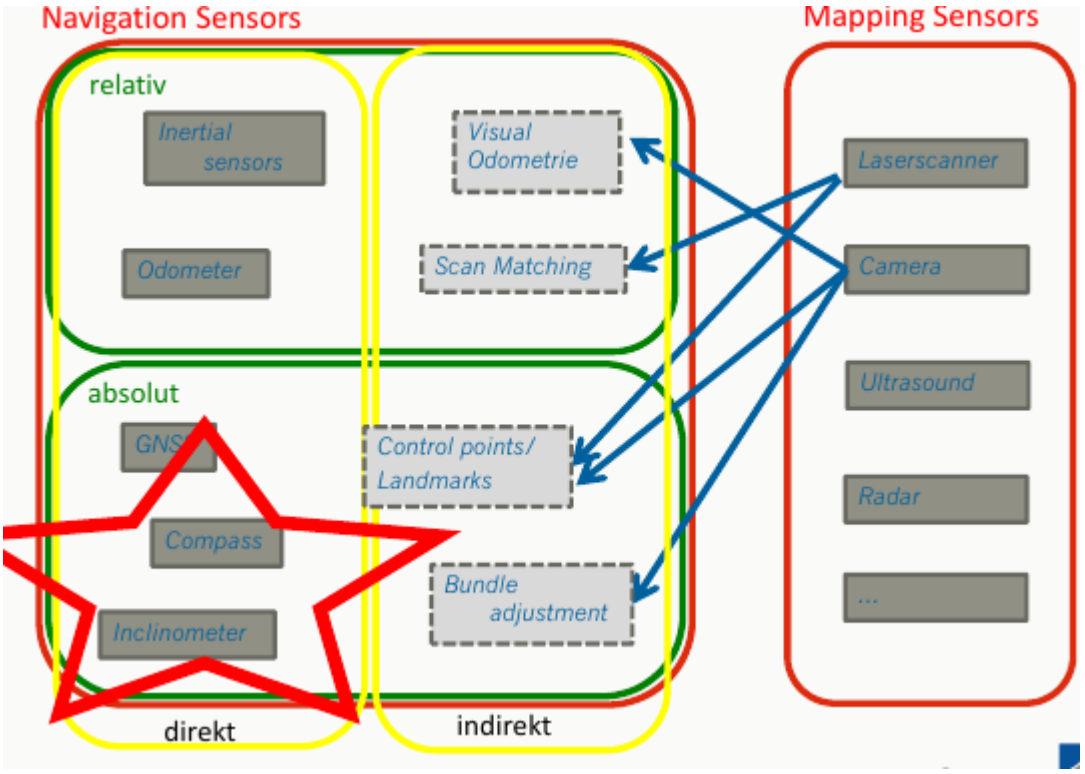
Ms05: Absolute Sensors



Initialisation & Alignment

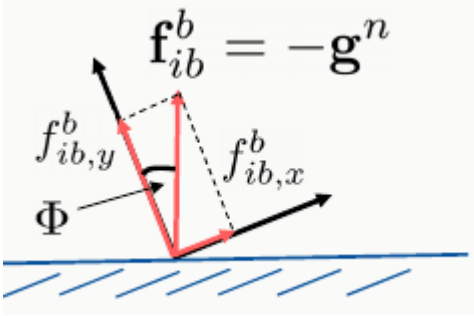
- **Position**:
  - Use absolute sensors (e.g. GNSS)
  - Start from a known position
- **Velocity**:
  - Use absolute sensor (e.g. GNSS)
  - Start in a static situation (v=0)
- **Orientation**:
  - Magnetic Compass
  - Inclinometer or accelerometer
  - Gyroscopic compass
  - Start from a known orientation

Orientation from vector observations

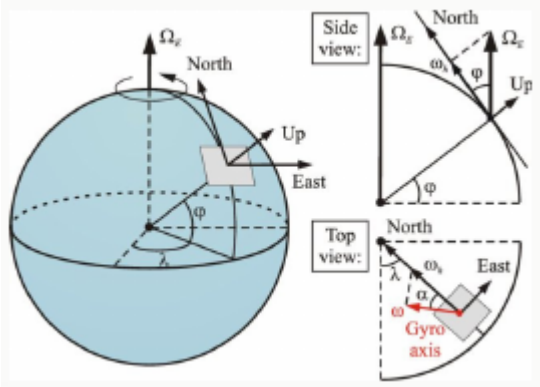
- Observe a vector $\boldsymbol{v}^e$, which is constant with respect to the Earth in the local coordinate system of the sensor $\boldsymbol{v}^b$
- Find rotation matrix $\boldsymbol{C}_b^e$ with $\boldsymbol{v}^e = \boldsymbol{C}_b^e \boldsymbol{v}^b$

- Not solvable with one vector observation (one angle remains underdetermined)
  - -> get only 2 angles OR use multiple vectors
- Options: gravity, magnetic field, Earth rotation rate

Accelerometer Levelling

- **In stationary systems**, roll and pitch can be derived from accelerometer
  - $\phi_{nb} = arctan(\frac{f^b_{ib,y}}{f^b_{ib,x}})$
  - 

Gyroscope Compassing

- Use Earth rotation rate to measure yaw angle
- Not possible with low cost IMUs, because Earth rate is below noise
- 

- Roll and Pitch from accelerometer leveling

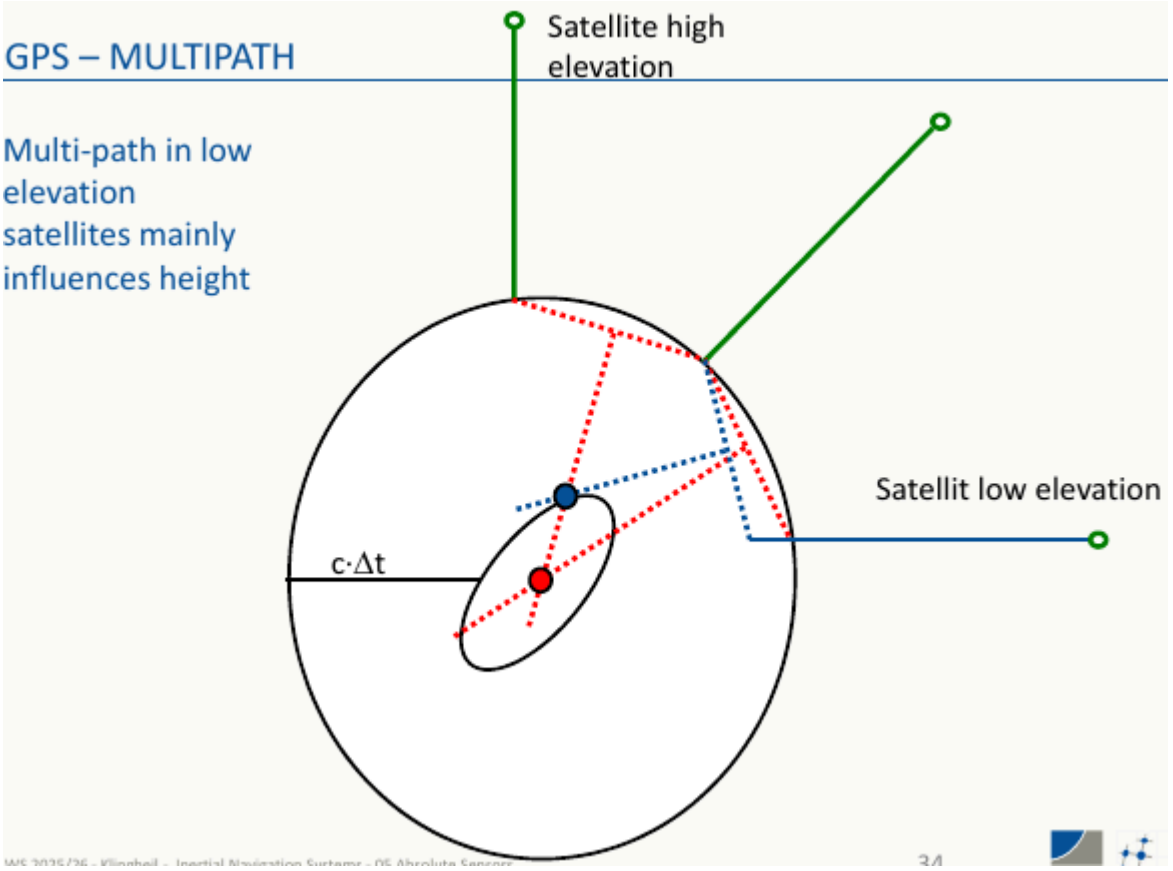
Magnetic Compassing

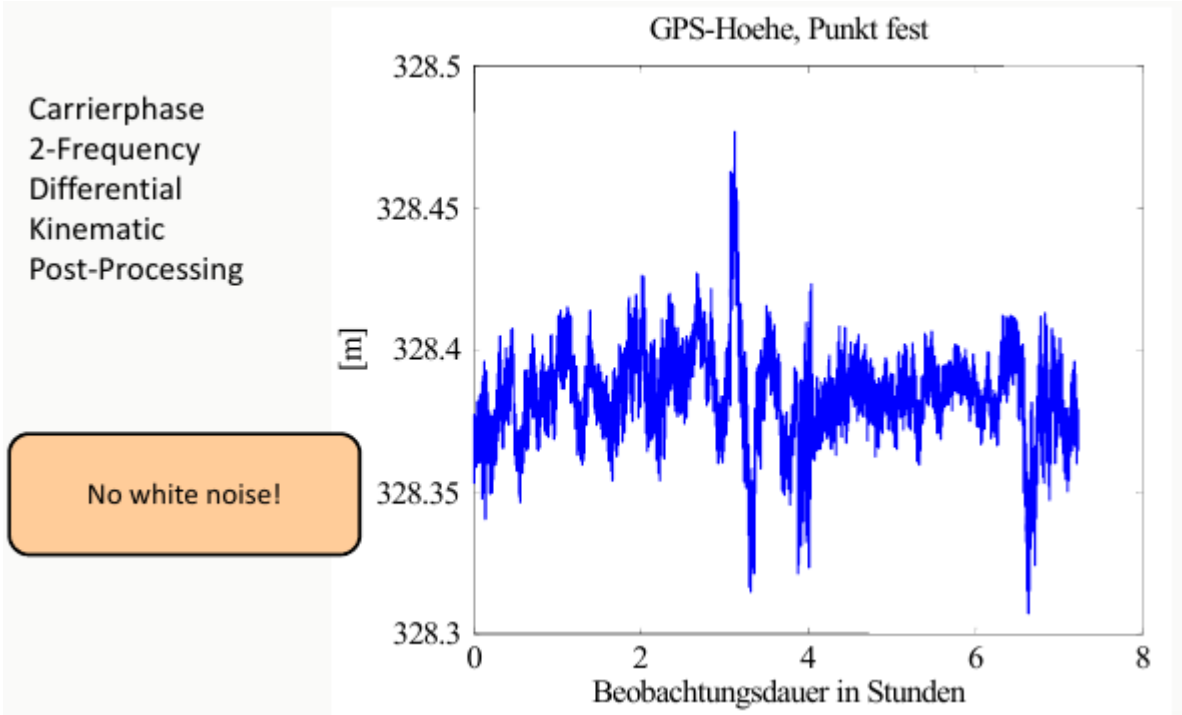
- Use compass to measure the 3D Earth magnetic field vector $\boldsymbol{m}^b_{mb}$
- Roll and Pitch from accelerometer leveling ($\boldsymbol{\Theta}_{nb}, \boldsymbol{\Phi}_{nb}$)
- Yaw is measured with respect to magnetic north! ($\Psi_{nb} = \Psi_{mb} + \alpha_{nE}$)

GNSS in Mobile Multi Sensor Systems

- Changing conditions
  - No averaging
  - Jumps
  - **Multipath**

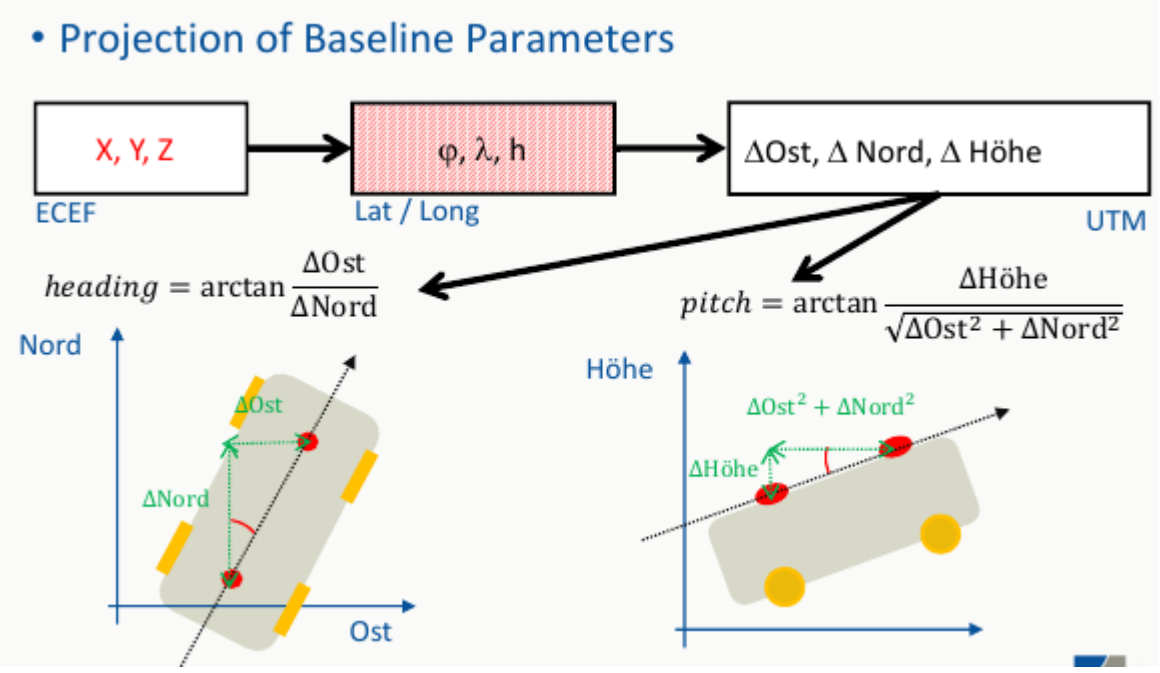
    - **Non-line-of-sight signals**
- Use **Carrier phase based relative positioning**
    - ambiguity fixing difficult for inaccurate float solutions
    - ambiguities have to be fixed again after every loss
- Use of **reference services**

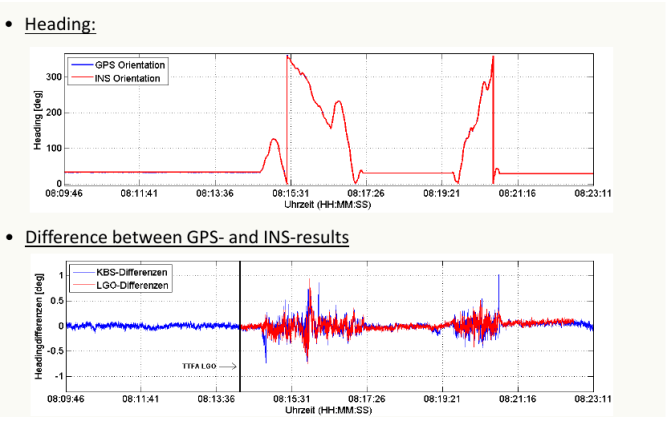


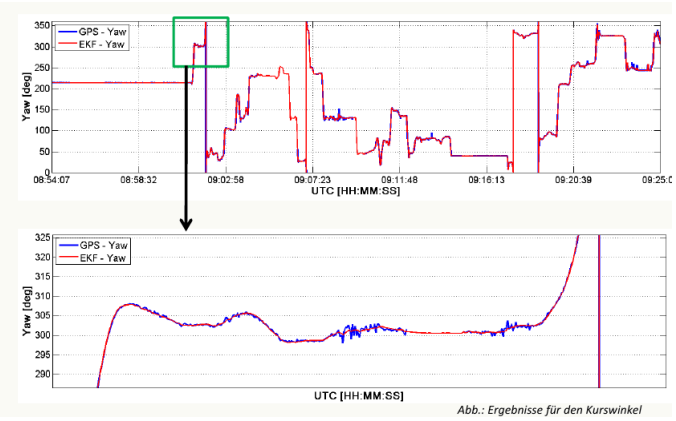


Orientation

- Using a sequence of position
  - Orientation vs. rotation
  - Track $\neq$ Heading
  - Orientation $\tilde{=}$ needs 2 epoches
  - -> Multi Antenna systems are better !
- Multi-Antenna System
  - Two Baselinevectors
    - Yaw- and Pitch-angle
    - Roll-angle
  - Accuracy depends on baseline length







GPS Velocity Determination

- Possible observations:
  - RTK-Position of a single antenna (differentiation)
  - Doppler observations generated by the receivers

- **Doppler measurmeent derived from carrier phase observations**

- $\dot{\phi}_k^j = \frac{\phi_{k+\Delta t}^j - \phi_{k-\Delta t}^j}{s\Delta t}$
- No ambiguity resolution necessary
- Results in 3D-velocity vector, if position and sat position are used
- Accuracy for 1-10Hz measurement rate
  - 1 mm/s - 2 cm/s